

Tutorial 1 - SPM (wk4 done)

Note: For calculation/practical questions, EACH assignment team shares ONE set of answer, i.e. if 4 assignment teams then only 4 sets of answers will be shared in the answer file. [Show all the workings clearly].

Q1 (prepared by Team 2)

Question

Question 1

Differentiate between a software project and a construction project by completing the Table 1 below. Include an example in each of your answers.

Table 1: Comparison of software project and construction project

	Software Project (example developing the Exam Timetable System)	Construction Project (example building a bridge)
Invisibility		
Flexibility		

Answer

	Software Project (example developing the Exam Timetable System)	Construction Project (example building a bridge)
Invisibility	Progress is not immediately visible E.g. Writing backend code for generating optimal exam schedules - no visual change yet	Progress is immediately visible E.g. Bridge foundation laid or pillars constructed can be clearly seen
Flexibility	Easier to change E.g. Updating the scheduling algorithm or changing user interface layout after testing	Difficult to change once it is completed E.g. Changing the bridge span or adding extra lanes after construction is costly and complex

Q2 (prepared by Team 1)

T-Shirts2U is a small company which supplies and manufactures customized t-shirts. However, its current system which is partially automated is unable to keep track of the orders and stocks efficiently. Besides that, it is unable to provide accurate information to the management on which items are more profitable and there have been mistakes in some of the order fulfillments. In order to solve these problems, the senior management of T-Shirts2U has requested you to develop a new online ordering system to increase the company annual sales by RM50,000. The new system to be developed has the following modules: Customer, Order carts, Online payment and Order fulfillment. This project must be completed within 3 months with an allocated budget of RM180,000. The project team consists of six software developers.

a. Identify and explain TWO (2) constraints of T-shirt2U project implementation. Explain your answer.

1. Time Constraint

- The project must be completed within 3 months. This is a strict time frame that limits how much work can be done and how detailed or advanced the system can be. Any delay in development, testing, or deployment could result in project failure or missing business objectives.

2. Budget Constraint

- The allocated budget is RM180,000. This. Budget restricts how many resources can be used (e.g., hiring more developers, purchasing software/tools, or conducting extensive testing). The team must manage cost carefully to avoid overspending while still delivering all required modules.

b. The outcome of a project can be classified as truly successful, partially successful or a total failure.

Identify and explain TWO (2) criteria that can be used to judge the above project whether it is a total failure.

1. Failure to Meet Business Objectives

- If the system fails to increase annual sales by RM50,000, it means the core business goal is not achieved. Despite technical completion, not achieving this target would classify the project as a failure in terms of return on investment.

2. Non-Completion Within Constraints

- If the project exceeds the 3-month timeline or goes over the RM180,000 budget, it fails to meet its key constraints. This could result in financial loss, stakeholder dissatisfaction, and the system potentially being or not deployed on time.

Q3 (prepared by Team 3)

Question 3

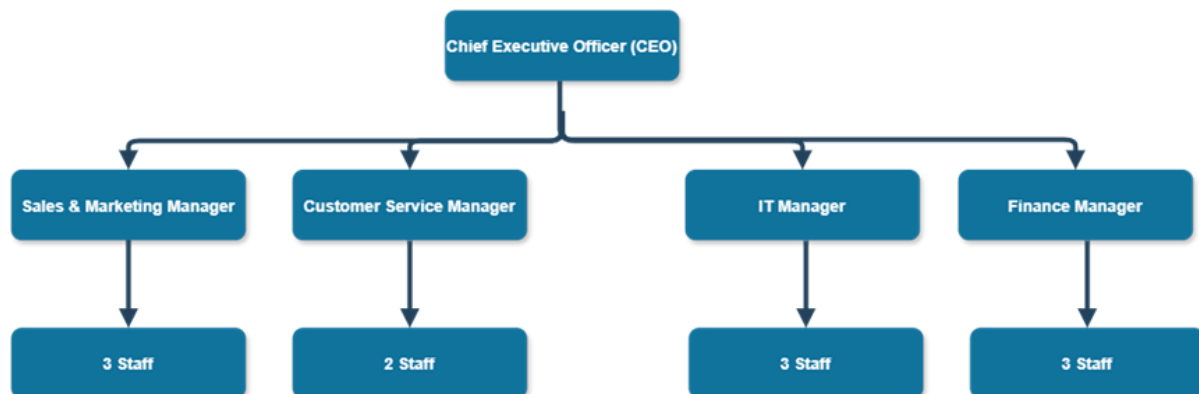
ABC Tech Company provides software consultancy services to business clients, it focuses on selling accounting software. The company consists of 4 different departments: sales and marketing, customer services, information technology (IT) and finance. Each staff member will be assigned to a department according to his or her area of specialization and expertise. The sales and marketing department consists of 3 staff, 2 staff work in the customer service department, while 3 staff work in the finance department and 3 experienced staff work in the IT department. Each department is supervised by a functional manager and each functional manager in turn reports to the Chief Executive Officer (CEO).

- Determine an appropriate organizational structure and create a corresponding diagram for this company.
- Based on part a), explain TWO (2) advantages of the selected organizational structure.

a.

Functional Organizational Structure

- Each department is independent and has a clear functional focus.
- It supports specialization and technical excellence.



b. Based on part a), explain TWO (2) advantages of the selected organizational structure.

1. Specialization and Expertise Development

- Employees are grouped by similar skills and tasks (e.g., IT staff under the IT department), which fosters expertise in specific domains. This leads to higher efficiency and quality output.

2. Clear Reporting and Supervision

- Each employee reports to a functional manager, ensuring well-defined roles and streamlined decision-making within each department.

Question

Question 4

You are a project manager of an online education management system (EMS) developed for a private college. After communicating with the staff for collecting requirements, the following scope is defined: the system will be used to manage communication between lecturer and students, uploading and downloading learning materials (such as tutorial questions, lecture notes, assignments specifications), making announcements, and assessing students' assignments. The modules of EMS can be modularized with a well-defined scope and high staff involvement. The project team focuses on the core features of the user interface design, assessment, communication, announcement and submission. A budget of RM150,000 has been approved for the project with a duration of 8 weeks.

Construct a *Work Breakdown Structure* (WBS) for the online education management system using an outline numbering format. Assume that you have chosen the *Waterfall software process model*.

Answer

- 1.0 Requirement analysis phase
 - 1.1 Requirements gathering
 - 1.1.1 Stakeholder meeting with college staff
 - 1.2 Requirements analysis
 - 1.3 Requirements documenting
 - 1.4 Requirements reviewing
- 2.0 System design phase
 - 2.1 Design system architecture
 - 2.2 Design system ui
 - 2.3 Design system database
 - 2.4 Define system modules
 - 2.4.1 Communication module
 - 2.4.2 Learning Materials module
 - 2.4.3 Announcements module
 - 2.4.4 Assessment module
 - 2.5 Prepare design documentation
- 3.0 System development phase
 - 3.1 Develop system modules
 - 3.1.1 Develop communication module
 - 3.1.2 Develop learning materials module
 - 3.1.3 Develop announcements module
 - 3.1.4 Develop assessment module
 - 3.2 Develop system ui
 - 3.3 Develop system database
 - 3.4 Integrate all system modules
- 4.0 Testing phase
 - 4.1 Unit testing
 - 4.2 Integration testing
 - 4.3 System testing
 - 4.4 User Acceptance Testing
- 5.0 Deployment phase
 - 5.1 Setting up hardware environment
 - 5.2 Software installation
 - 5.3 Database migration
 - 5.4 Deploy system on live server
 - 5.5 User training
- 6.0 Maintenance phase

- 6.1 Monitor system performance
- 6.2 Develop and deploy bug fixes
- 6.3 Collect user feedback for improvements

Q5 (prepared by Team ?)

Team 1

Year	Project ABC Cash Flow(RM)	Discount Factor	Present value
0	- 50,000	1.0000	- 50,000
1	20,000	0.8929	17,858
2	20,000	0.7972	15,944
3	20,000	0.7118	14,236
4	20,000	0.6355	12,710
Net Profit	40,000		NPV : 17,103
ROI	20%		

$$ROI = \frac{(40,000 \div 4)}{50,000} \times 100 = 20\%$$

$$\text{Discount Factor : } Y1 = 1 / (1 + 0.12)^1 \\ = 0.8929$$

$$NPV = - 50,000 + 17,858 + 15,944 + 14,236 + 12,710 \\ = 17,103$$

$$Y2 = 1 / (1 + 0.12)^2 \\ = 0.7972$$

$$Y3 = 1 / (1 + 0.12)^3 \\ = 0.7118$$

∴ The project is attractive, because NPV > 0 when discount rate is 12% and its ROI is high (20%).

$$Y4 = 1 / (1 + 0.12)^4 \\ = 0.6355$$

Team 2

Question 5

Return on Investment (ROI), Net Present Value (NPV) and many other financial techniques can be used to evaluate and compare the risks and rewards of different projects. They help project managers to decide whether a project is worthwhile to pursue or not. Table 1 illustrates the cash flow of Project ABC. The cash flow takes place at the end of each year.

Table 1: Cash flow of Project ABC

Year	Project X
0	-50,000
1	20,000
2	20,000
3	20,000
4	30,000

Using a 12% discount rate, calculate the Net Present Value (NVP) and Return on Investment (ROI) for this Project. Based on your calculations, analyse whether the project is attractive, or should be rejected. Explain your answer.

Note:

- Show your workings in calculation.
- Negative values represent costs.
- Discount factors round up to 4 decimal points. All other figures round up to the nearest integer.
- ROI in percentage, rounds up to 2 decimal points.

NPV = Present Value of Cash Inflows - Initial Investment

Year	Cash Flow (RM)	Discount Factor	PV (RM)
0	-50,000	$1 / (1 + 0.12)^0 = 1.0000$	$-50,000 * 1.0000 = -50,000$
1	20,000	$1 / (1 + 0.12)^1 = 0.8929$	$20,000 * 0.8929 = 17,858$
2	20,000	$1 / (1 + 0.12)^2 = 0.7972$	$20,000 * 0.7972 = 15,944$
3	20,000	$1 / (1 + 0.12)^3 = 0.7118$	$20,000 * 0.7118 = 14,236$
4	30,000	$1 / (1 + 0.12)^4 = 0.6355$	$30,000 * 0.6355 = 19,065$

$$\begin{aligned}\text{NPV} &= -50,000 + 17,858 + 15,944 + 14,236 + 19,065 \\ &= \text{RM } 17,103\end{aligned}$$

Return on Investment (ROI)

$$\begin{aligned}\text{Net Profit} &= \text{Total Inflows} - \text{Initial Investment} \\ &= (20,000 + 20,000 + 20,000 + 30,000) - 50,000 \\ &= 90,000 - 50,000 \\ &= \text{RM } 40,000\end{aligned}$$

Method 1

$$\begin{aligned}\text{ROI} &= (\text{NPV} / \text{Discount Cost}) * 100\% \\ &= 17103 / 50000 * 100\% \\ &= 34.21\%\end{aligned}$$

Method 2

$$\begin{aligned}\text{ROI} &= (\text{Average Annual Profit} / \text{Total Investment}) * 100\% \\ &= ((40,000 / 4) / 50,000) * 100\% \\ &= 20.00\%\end{aligned}$$

Based on the calculation of ROI and NPV, Project X is attractive. This is because the ROI is 20%, which means the company gets to earn 20% from their investment within 4 years. Besides, Project X NPV is positive. A positive NPV indicates the project is profitable [PV of cash inflows (benefits) > PV of cash outflows (costs) over the 4-year period]. Project X has achieved the 12% target rate of return. This result shows that the project is expected to be profitable and financially worthwhile.

Team 3

5.

Year	Project X	Discount factor	Present Value
0	-50,000	$\frac{1}{(1+0.12)^0} = 1$	-50,000
1	20,000	$\frac{1}{(1+0.12)^1} = 0.8929$	$20,000 \times 0.8929 = 17,858$
2	20,000	$\frac{1}{(1+0.12)^2} = 0.7972$	$20,000 \times 0.7972 = 15,944$
3	20,000	$\frac{1}{(1+0.12)^3} = 0.7118$	$20,000 \times 0.7118 = 14,236$
4	30,000	$\frac{1}{(1+0.12)^4} = 0.6355$	$30,000 \times 0.6355 = 19,065$
	40,000		17,103

$$NPV = \sum \text{Present Value} = \text{RM}17,103$$

$$ROI = \frac{\text{average annual profit}}{\text{total investment}} \times 100$$

$$= \frac{40,000/4}{50,000} \times 100$$

$$= \frac{10,000}{50,000} \times 100$$

$$= 20\%$$

$\therefore$ NPV and ROI > 0, so is attractive

Team 4**Question 5**

Year	Project X	Discount Factor	Present Value
0	-50,000	$1/(1+0.12)^0 = 1.0000$	$-50,000 * 1.0000 = -50,000$
1	20,000	$1/(1+0.12)^1 = 0.8929$	$20,000 * 0.8929 = 17,858$
2	20,000	$1/(1+0.12)^2 = 0.7972$	$20,000 * 0.7972 = 15,944$
3	20,000	$1/(1+0.12)^3 = 0.7118$	$20,000 * 0.7118 = 14,236$
4	30,000	$1/(1+0.12)^4 = 0.6355$	$30,000 * 0.6355 = 19,065$
Net Profit	$-50,000 + 20,000 + 20,000 + 20,000 + 30,000 = 40,000$		
NPV	$-50,000 + 17,858 + 15,944 + 14,236 + 19,065 = 17,103$		
ROI(Method 1)	$(40,000/4)/50,000*100 = 20\%$		
ROI(Method 2)	$NPV/Discounted\ Cost * 100\% = (17,103/50,000)*100\% = 34.21\%$		

Based on the calculation of ROI and NPV, Project X is attractive. This is because the ROI is 20%, which means the company gets to earn 20% from their investment within 4 years. Besides, Project X has reflected positive NPV which indicates the project is profitable as the present value of inflows (benefits) is larger than the present value of outflows (cost) over the 4-year period. Project X has achieved the 12% target rate of return. This result shows that the project is expected to be profitable and financially worthwhile.

Q6 (prepared by Team ?)

Team 1

Discount Rate: 10% for both projects					
Project X	Year 0	Year 1	Year 2	Year 3	Total
Costs	100,000	50,000	40,000	30,000	-
Discount Factor	1	0.9091	0.8264	0.7513	
Discounted Costs	100,000	45,455	33,056	22,539	201,050
Benefits	0	100,000	120,000	100,000	-
Discount Factor	1	0.9091	0.8264	0.7513	
Discounted Benefits	0	90,910	99,168	75,130	265,208
NPV	64,158				
ROI: Use discounted value	31.91%				

Project Y	Year 0	Year 1	Year 2	Year 3	Total
Costs	100,000	70,000	60,000	20,000	-
Discount Factor	1	0.9091	0.8264	0.7513	
Discounted Costs	100,000	63,637	49,584	15,026	228,247
Benefits	0	100,000	150,000	90,000	-
Discount Factor	1	0.9091	0.8264	0.7513	
Discounted Benefits	0	90,910	123,960	67,617	282,487
NPV	54,240				
ROI:	23.76%				

Comment:

$$\text{Discount Factor : Year 1} = 1 \div (1 + 0.10)^1 = 0.9091$$

$$\text{Year 2} = 1 \div (1 + 0.10)^2 = 0.8264$$

$$\text{Year 3} = 1 \div (1 + 0.10)^3 = 0.7513$$

Project X

$$\text{NPV} = \text{RM}265,208 - \text{RM}201,050 = \text{RM}64,158$$

$$\text{ROI} = (64,158 / 201,050) \times 100 = 31.91\%$$

Project Y

$$\text{NPV} = \text{RM}282,487 - \text{RM}228,247 = \text{RM}54,240$$

$$\text{ROI} = (54,240 / 228,247) \times 100 = 23.76\%$$

$\therefore$ Project X is profitable than project Y, because Project X ROI (31.91%) and NPV is higher than Project Y ROI (23.76%).

Team 2

Question 6

Using a 10% discount rate, calculate the Net Present Value (NPV) and Return on investment (ROI) for Project X and Project Y. Based on your calculations, suggest the most profitable project. Explain your answer.

Show your calculation to justify your answer. Discount factors round up to 4 decimal points.

Discount Rate: 10% for both projects					
Project X	Year 0	Year 1	Year 2	Year 3	Total
Costs	100,000	50,000	40,000	30,000	-
Discount Factor	1				
Discounted Costs	100,000				
Benefits	0	100,000	120,000	100,000	-
Discount Factor	1				
Discounted Benefits	0				
NPV					
ROI:					
Use discounted value					

Project Y	Year 0	Year 1	Year 2	Year 3	Total
Costs	100,000	70,000	60,000	20,000	-
Discount Factor	1				
Discounted Costs	100,000				
Benefits	0	100,000	150,000	90,000	-
Discount Factor	1				
Discounted Benefits	0				
NPV					
ROI:					

Comment:

Discount Rate: 10% for both projects					
Project X	Year 0	Year 1	Year 2	Year 3	Total
Costs	100,000	50,000	40,000	30,000	-
Discount Factor	$1 / (1 + 0.10)^0 = 1.0000$	$1 / (1 + 0.10)^1 = 0.9091$	$1 / (1 + 0.10)^2 = 0.8264$	$1 / (1 + 0.10)^3 = 0.7513$	
Discounted Costs	$100,000 * 1.000 = 100,000$	$50,000 * 0.9091 = 45,455$	$40,000 * 0.8264 = 33,056$	$30,000 * 0.7513 = 22,539$	$100,000 + 45,455 + 33,056 + 22,539 = 201,050$
Benefits	0	100,000	120,000	100,000	-

Discount Factor	$1 / (1 + 0.10)^0 = 1.0000$	$1 / (1 + 0.10)^1 = 0.9091$	$1 / (1 + 0.10)^2 = 0.8264$	$1 / (1 + 0.10)^3 = 0.7513$	
Discounted Benefits	$0 * 1.000 = 0$	$100,000 * 0.9091 = 90,910$	$120,000 * 0.8264 = 99,168$	$100,000 * 0.7513 = 75,130$	$0 + 90,910 + 99,168 + 75,130 = 265,208$
NPV	Total Discount Benefit - Total Discount Cost $265,208 - 201,050 = 64,158$				
ROI	$(NPV / \text{Discount Cost}) * 100\%$ $(64,158 / 201,050) * 100 = 31.91\%$				

Use discounted value

Project Y	Year 0	Year 1	Year 2	Year 3	Total
Costs	100,000	70,000	60,000	20,000	-
Discount Factor	$1 / (1 + 0.10)^0 = 1.0000$	$1 / (1 + 0.10)^1 = 0.9091$	$1 / (1 + 0.10)^2 = 0.8264$	$1 / (1 + 0.10)^3 = 0.7513$	
Discounted Costs	$100,000 * 1.000 = 100,000$	$70,000 * 0.9091 = 63,637$	$60,000 * 0.8264 = 49,584$	$20,000 * 0.7513 = 15,026$	$100,000 + 63,637 + 49,584 + 15,026 = 228,247$
Benefits	0	100,000	150,000	90,000	-
Discount Factor	$1 / (1 + 0.10)^0 = 1.0000$	$1 / (1 + 0.10)^1 = 0.9091$	$1 / (1 + 0.10)^2 = 0.8264$	$1 / (1 + 0.10)^3 = 0.7513$	
Discounted Benefits	$0 * 1.000 = 0$	$100,000 * 0.9091 = 90,910$	$150,000 * 0.8264 = 123,960$	$90,000 * 0.7513 = 67,617$	$0 + 90,910 + 123,960 + 67,617 = 282,487$
NPV	Total Discount Benefit - Total Discount Cost $282,487 - 228,247 = 54,240$				
ROI (Use discounted)	$(NPV / \text{Discount Cost}) * 100\%$				

value)	$(54240 / 228,247) * 100 = 23.76\%$
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Project X NPV is positive. A positive NPV indicates the project is profitable [PV of cash inflows (benefits) > PV of cash outflows (costs) over the 4-year period]. Project X NPV is > Project Y (PX is profitable than PY)

Team 3

6.

answer.

Show your calculation to justify your answer. Discount factors round up to 4 decimal points.

Discount Rate: 10% for both projects					
Project X	Year 0	Year 1	Year 2	Year 3	Total
Costs	100,000	50,000	40,000	30,000	-
Discount Factor	1	$1/(1+0.10)^1 = 0.9091$	$1/(1+0.10)^2 = 0.8264$	$1/(1+0.10)^3 = 0.7513$	
Discounted Costs	100,000	45455	33056	22539	201050
Benefits	0	100,000	120,000	100,000	-
Discount Factor	1	0.9091	0.8264	0.7513	
Discounted Benefits	0	90910	99168	75130	265208
NPV	$-100000 + 45455 + 66112 + 52591 = RM 64158$				
ROI:	$(64158 / 201050) \times 100\% = 31.91\%$				
Use discounted value					

Project Y	Year 0	Year 1	Year 2	Year 3	Total
Costs	100,000	70,000	60,000	20,000	-
Discount Factor	1	0.9091	0.8264	0.7513	
Discounted Costs	100,000	63637	49584	15026	RM 228247
Benefits	0	100,000	150,000	90,000	-
Discount Factor	1	0.9091	0.8264	0.7513	
Discounted Benefits	0	90910	123960	67617	RM 282487
NPV	$-100000 + 27273 + 74376 + 52591 = RM 54240$				
ROI:					

Comment:

$\left(\frac{54240}{228247} \right) \times 100 = 23.76\% \therefore \text{choose project X, because ROI is higher.}$

Team 4

Question 6

Discount Rate : 10% for both project					
Project X	Y0	Y1	Y2	Y3	Total
Costs	100,000	50,000	40,000	30,000	-
Discount Factor	$1/(1+0.1)^0 = 1.0000$	$1/(1+0.1)^1 = 0.9091$	$1/(1+0.1)^2 = 0.8264$	$1/(1+0.1)^3 = 0.7513$	-
Discounted Costs	$100,000 * 1.0000 = 100,000$	$50,000 * 0.9091 = 45,455$	$40,000 * 0.8264 = 33,056$	$30,000 * 0.7513 = 22,539$	$100,000 + 45,455 + 33,056 + 22,539 = 201,050$
Benefits	0	100,000	120,000	100,000	-
Discount Factor	1	0.9091	0.8264	0.7513	-
Discounted Benefits	0	$100,000 * 0.9091 = 90,910$	$120,000 * 0.8264 = 99,168$	$100,000 * 0.7513 = 75,130$	$0 + 90,910 + 99,168 + 75,130 = 265,208$
NPV	$265,208 - 201,050 = 64,158$				
ROI: Use discounted value	$(NPV/Discounted\ Cost) * 100\%$ $= (64,158/201,050) * 100\%$ $= 31.91\%$				

Project Y	Y0	Y1	Y2	Y3	Total
Costs	100,000	70,000	60,000	20,000	-
Discount Factor	1.0000	$1/(1+0.1)^1 = 0.9091$	$1/(1+0.1)^2 = 0.8264$	$1/(1+0.1)^3 = 0.7513$	-
Discounted Costs	$100,000 * 1.0000 = 100,000$	$70,000 * 0.9091 = 63,637$	$60,000 * 0.8264 = 49,584$	$20,000 * 0.7513 = 15,026$	$100,000 + 63,637 + 49,584 + 15,026 = 228,247$
Benefits	0	100,000	150,000	90,000	-
Discount Factor	1	0.9091	0.8264	0.7513	-

Discounted Benefits	0	100,000 * 0.9091 = 90,910	120,000 * 0.8264 = 99,168	90,000 * 0.7513 = 67,617	0 + 90,910 + 99,168 + 67,617 =257,695
NPV	257,695 - 228,247=29,448				
ROI: Use discounted value	(NPV/Discounted Cost) * 100% = (29,448/228,247) * 100% = 12.90%				

Comment: The most profitable project is Project X. The NPV of Project X is positive which is 64,158 indicates the project is profitable compare with Project Y which is 29,448 is less than 64,158.

MidTerm Test (Session 2025/01) Sample

BAIT3153 Software Project Management (202501) **Test** 1

Student Name: _____ (R____ Group: __) Student Id: _____ Tutor: _____

Total Marks: _____ X 4 = _____ / 100

INSTRUCTION: Answer all the questions. [Total: 25 marks]

You have joined Wemax company which provides software solutions service. Jacky, CEO of the company has assigned you to manage a project to develop an E-Schedule System (ESS) for rail services. A budget of RM170,000 has been approved for the project. The project team consists of members with expertise in specific knowledge areas and experienced staff from the Information Technology department who will contribute to this project.

Answer the following questions based on the above case scenario.

1. Net Present Value (NPV) is a financial technique used to compare the profitability of different projects and timing of the cash flows produced at the end of each year. Table 1 illustrates the cash flow of Project ESS.

Table 1: Cash flow of Project ESS

Year	Cash flow
0	-170,000
1	110,000
2	80,000
3	40,000

Calculate the following values for this project, show working in your calculation.

a) Net Present Value (NPV) using a discount rate of 12%.

Note - negative value represent cost.
 - discount rate must be rounded up to four decimal places.
 - NPV and present value round up to the nearest integer.

(9 marks)

Answer:

Discount Rate: 12%					
Project ESS	Year 0	Year 1	Year 2	Year 3	Total

b) Return on Investment (ROI) in percentage, round up to two decimal places. (2 marks)

Answer:

BAIT3153 Software Project Management (202501) Test

2

2. Table 2 shows an example of project tasks, duration and its predecessor.

Table 2: Project tasks, duration and its predecessor.

Task ID	Task Name	Duration (in week)	Predecessor
1	Meet stakeholders for discussion	1	None
2	Requirement gathering	1	1
3	Data modeling	1	2
4	Process modeling	2	2
5	Application generation	2	3, 4
6	System testing	2	5

a) Draw a network diagram based on the information provided in Table 2. (7 marks)

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Answer:

b) Based on the answer given in 2.a), answer the following questions.

- Identify the critical path of the project.
- Estimate the total duration to complete the project.

(2 marks)

(1 mark)

3. Given below are two organizational structures:

- Functional
- Project

Choose an appropriate organizational structure from the options above for the ESS's project team and provide an explanation for your choice.

(4 marks)

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Answer: